



Islamic University of Technology
Department of Computer Science and Engineering

Lab 7: Dynamic Programming

CSE 4404: Algorithms Lab
Summer 2024-25

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Task A. Tribonacci Staircase

Time Limit: 1 second | Memory Limit: 256 MB

A frog wants to climb n steps. From any step, it can jump exactly 1, 2, or 3 steps forward. Count the number of distinct ways the frog can reach step n starting from step 0.

Input Format

A single integer n .

Output Format

Print the number of distinct ways to reach step n .

Constraints

- $1 \leq n \leq 70$

Examples

Sample Input	Sample Output
4	7
5	13

Task B. Frog Steps

Time Limit: 1 second | Memory Limit: 256 MB

A frog wants to cross a sequence of n stones numbered 1 to n . Stone i has a landing cost c_i . The frog starts on stone 1 and must reach stone n . From stone i , it may jump to stone $i + 1$ or stone $i + 2$. The total cost is the sum of landing costs of all stones visited. Compute the minimum total cost to reach stone n .

Input Format

The first line contains an integer n . The second line contains n integers $c_1, c_2, \dots, c_n$.

Output Format

Print the minimum total cost to reach stone n .

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq c_i \leq 10^6$

Examples

Sample Input	Sample Output
5 2 3 1 5 2	5
6 1 5 3 2 4 1	7

Task C. Frog K Steps

Time Limit: 1 second | Memory Limit: 256 MB

There are n stones numbered 1 to n . Stone i has height h_i . A frog starts on stone 1 and wants to reach stone n . From stone i , it can jump to any stone j where $i < j \leq i + k$. The cost of jumping from stone i to stone j is $|h_i - h_j|$. Find the minimum total cost to reach stone n .

Input Format

The first line contains two integers n and k . The second line contains n integers $h_1, h_2, \dots, h_n$.

Output Format

Print the minimum total cost to reach stone n .

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq k \leq n - 1$
- $1 \leq h_i \leq 10^9$

Examples

Sample Input	Sample Output
5 3 10 30 40 20 10	20
6 2 1 5 3 2 4 2	3

Task D. House Robber

Time Limit: 1 second | Memory Limit: 256 MB

There are n houses in a row, each containing a non-negative amount of money a_i . A robber wants to maximize the total money stolen, but cannot rob two adjacent houses (doing so triggers an alarm). Find the maximum amount the robber can steal.

Input Format

The first line contains an integer n . The second line contains n integers $a_1, a_2, \dots, a_n$.

Output Format

Print the maximum amount the robber can steal.

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq a_i \leq 10^9$

Examples

Sample Input	Sample Output
4 1 2 3 1	4
5 2 7 9 3 1	12

Marks Distribution

Task	Marks
Task A	20%
Task B	25%
Task C	25%
Task D	30%

Practice Problems

Problem	Link
Frog 1	AtCoder EDU DP A: https://atcoder.jp/contests/dp/tasks/dp_a
Frog 2 (K Steps)	AtCoder EDU DP B: https://atcoder.jp/contests/dp/tasks/dp_b
Minimizing Coins	CSES: https://cses.fi/problemset/task/1634
Coin Combinations I	CSES: https://cses.fi/problemset/task/1631
Maximum Sum Increasing Subsequence	GeeksforGeeks: https://www.geeksforgeeks.org/maximum-sum-increasing-subsequence-dp-14/
Counting Staircase Ways	HackerRank / practice DP exercise (use step sizes 1 and 2)